Components

This project uses a combination of mechanical and electronic components to detect abnormal force on the ceiling fan rod and generate immediate safety alerts.

1. **ESP32 Microcontroller**

The ESP32 serves as the main controller of the system. It continuously monitors the input from the roller limit switch and executes the programmed safety actions when an abnormal condition is detected.

Features

- Built-in Wi-Fi connectivity
- Dual-core processor
- Multiple GPIO pins
- Low power consumption
- Supports IoT applications

Purpose in Project

- Reads the limit switch status
- Controls the relay and buzzer
- Sends Telegram alerts through Wi-Fi

2. Roller Limit Switch

A roller-type limit switch is used as the primary sensing device.

Purpose in Project

- Detects downward displacement of the fan rod
- Acts as the trigger mechanism
- Sends a signal to ESP32 when activated

Advantages

- Simple operation
- Reliable triggering
- Low cost
- Easy installation

3. Relay Module

The relay module acts as an electrically controlled switch

Purpose in Project

- Disconnects power to the ceiling fan
- Prevents further operation during emergency conditions

Features

- Electrical isolation
- Fast switching
- Suitable for AC load control

4. Buzzer

The buzzer provides an audible alert when the system detects abnormal force.

Purpose in Project

- Alerts nearby people
- Enables immediate local response
- Acts as a warning indicator

5. Spring Mechanism

The spring mechanism allows controlled axial movement of the fan rod.

Purpose in Project

- Enables downward displacement under excessive load
- Helps activate the sensing mechanism
- Provides mechanical support to the safety system

Benefits

- Simple design
- Reliable operation
- Minimal maintenance

6. Fan Rod Clamp Assembly

The clamp is attached to the fan rod and positioned near the roller limit switch.

Purpose in Project

- Transfers rod movement to the switch
- Maintains proper alignment
- Ensures accurate triggering

7. Power Supply

A regulated power supply is used to operate the electronic components.

Purpose in Project

- Powers ESP32
- Powers relay module
- Powers buzzer
- Ensures stable system operation

System Architecture

Fan Rod Movement

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Spring Mechanism

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Roller Limit Switch

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ESP32 Controller

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Relay + Buzzer

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Telegram Alert

Summary

The SafeSpin system combines mechanical sensing, embedded electronics, and wireless communication to provide a cost-effective and reliable safety solution. Each component plays a critical role in detecting abnormal conditions and generating rapid alerts for timely intervention.